Distributed Systems: Overview

Material adapted from Lectures by Indy Gupta (IG) of UIUC

What This Course is About

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- Distributed Systems
- How to Design Algorithms for them
- How to Design Such Systems
- How they work in real life
- How to build real distributed systems

Can you name some examples of Distributed Systems?

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- Client-Server (NFS)
- The Web
- The internet
- A wireless network
- DNS
- Gnutella or BitTorrent (peer to peer overlays)
- A “cloud”, e.g., Amazon EC2/S3, Microsoft Azure
- A datacenter, e.g., Google datacenter, AWS
- …

Can you name some examples of Distributed Systems?

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- Dropbox, Google Drive etc.
  - Client-Server
- Bitcoin
  - Distributed Ledger

A working definition for us

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A distributed system is a collection of entities, each of which is autonomous, programmable, asynchronous and failure-prone, and which communicate through an unreliable communication medium.

- Entity=a process on a device (PC, PDA)
- Communication Medium=Wired or wireless network
- Our interest in distributed systems involves
  - design and implementation, maintenance, algorithmics

Gnutella Peer to Peer System

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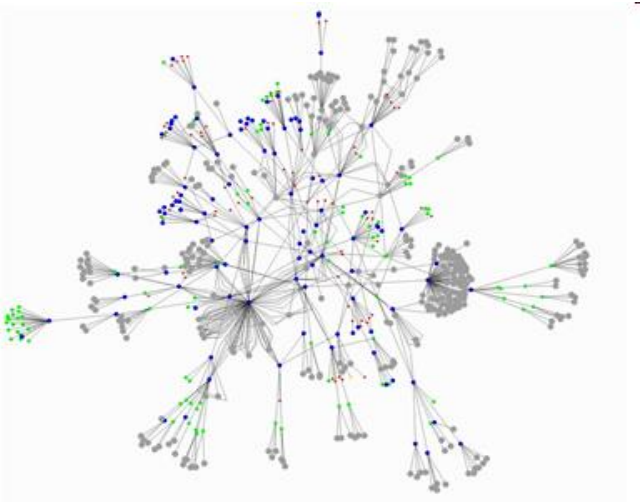

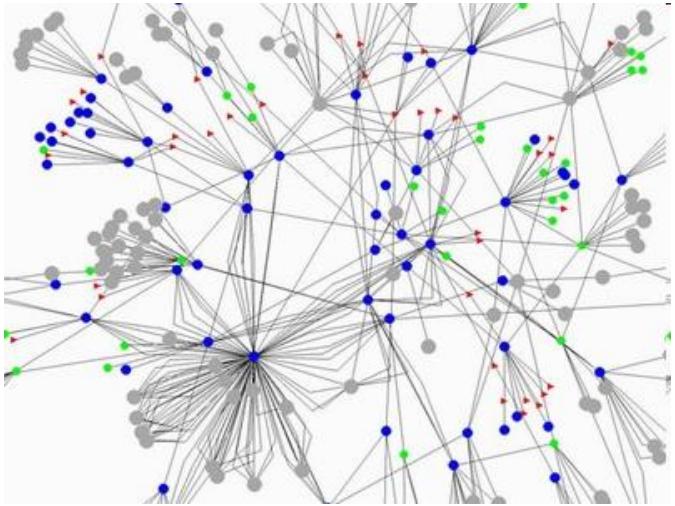

What are the "entities" (nodes)?

What is the communication medium (links)?

Source: GnuMap Project

Web Domains

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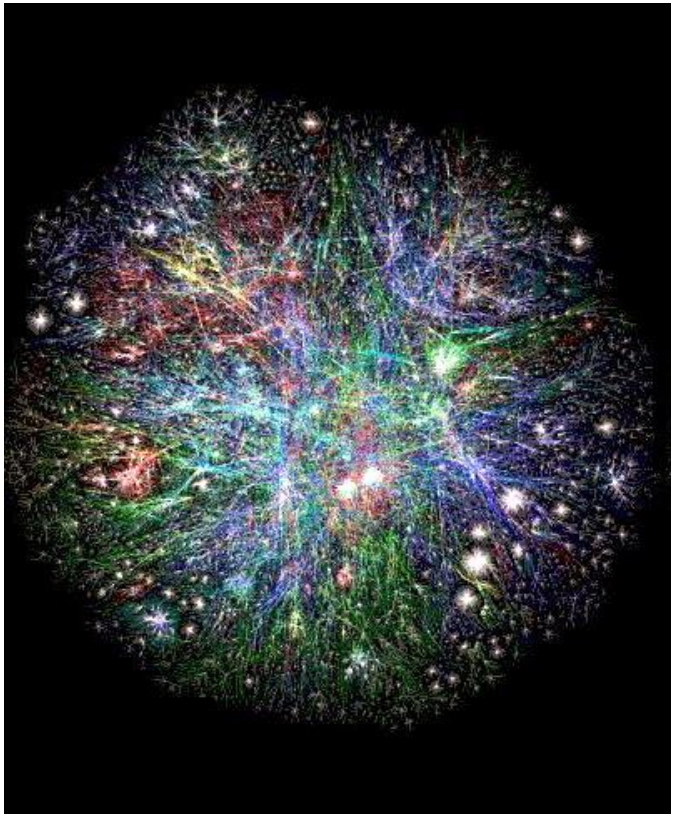
What are the "entities" (nodes)?

What is the communication medium (links)?



Source: http://www.vlib.us/web/worldwideweb3d.html

Datacenter

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What are the "entities" (nodes)?

What is the communication medium (links)?

"Important" Distributed Systems Issues

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- Asynchrony: No global clock; no single global notion of the correct time
- Reliability (failure-prone, unreliable): Unpredictable failures of components: lack of response may be due to
  - failure of a network component,
  - network path being down, or
  - a computer crash
- Scalability: Large numbers of hosts: 2 to several million
- Network Heterogeneity:
  - Highly variable bandwidth: from 16Kbps (slow modems or Google Balloon) to Gbps (Internet2) to Tbps (in between DCs of same big company)
- Latency variability: Possibly large and variable latency
  - few ms to several seconds

Many Interesting Design Problems

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- Real distributed systems
  - Cloud Computing, Peer to peer systems, Hadoop, key-value stores/NoSQL, distributed file systems, sensor networks, measurements, graph processing, stream processing, distributed ledgers (Bitcoin) …
- Classical Problems
  - Failure detection, Asynchrony, Snapshots, Multicast, Consensus, Mutual Exclusion, Election, …
- Concurrency
  - RPCs, Concurrency Control, Replication Control, Paxos, …
- Security
  - ACLs, Capabilities, …
- Others…

Typical Distributed Systems Design Goals

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- Heterogeneity – can the system handle a large variety of types of PCs and devices?
- Robustness – is the system resilient to host crashes and failures, and to the network dropping messages?
- Availability – are data+services always there for clients?
- Transparency – can the system hide its internal workings from the users? (warning: term means the opposite of what the name implies!)

Typical Distributed Systems Design Goals

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- Concurrency – can the server handle multiple clients simultaneously?
- Efficiency – is the service fast enough? Does it utilize 100% of all resources?
- Scalability – can it handle 100 million nodes without degrading service? (nodes=clients and/or servers) How about 6 B? More?
- Security – can the system withstand cyber attacks?
- Openness – is the system extensible?

“Important” Issues

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If you’re already worried that the list of topics we’ve discussed so far has been perplexing…

You’re right!

It was meant to be (perplexing)

The Goal for the Rest of the Course: see enough examples and learn enough concepts so these topics and issues will make sense

“Concepts”?

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Which of the following inventions do you think is the most important?

1. Car
2. Wheel
3. Bicycle

“What lies beneath?” Concepts!

So, what is the course about?

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- Intended to be first course in distributed systems at OSU
  - Mix of motivation, concepts, and mechanisms
  - Covers concepts and principles underlying distributed system design
    - Breadth first look

Aims

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- Introduce you to distributed systems
  - Challenges, Concepts, Principles Building Blocks
- Insight into designing and building distributed systems
- Specific Topics
  - Time, Ordering, Synchronization, Failure Detection and Membership, Leader Election, Agreement and Consensus, Gossiping and Multicast, Fault-tolerance etc.
- Specific Systems/Paradigms
  - Map Reduce
  - Raft
  - Bitcoin (time permitting)

Learning Objectives

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At the end of this course:

- have an understanding of and appreciation for the challenges underlying the design and building of distributed systems
- know basic building blocks of distributed systems
- hopefully be able to design and build simple distributed systems

Class Overview - Resources

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Class page: Canvas

Discussion: Canvas

Academic Honesty

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Review student conduct code regarding academic dishonesty

http://studentlife.oregonstate.edu/studentconduct/offenses-0

Expectations for exams, projects, and homeworks

When in doubt, ask!

Questions?

Can you name some examples of Operating Systems?

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…

Linux

Win XP, Vista 7/8 Windows 10

Unix FreeBSD, Mac OSX

iOS, Android, RTOS

OS/2, PalmOS, WinCE, TinyOS

…

What is an Operating System?

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- User interface to hardware (device driver)
- Provides abstractions (processes, file system)
- Resource manager (scheduler)
- Means of communication (networking)
- …

FOLDOC definition

(FOLDOC = Free On-Line Dictionary of Computing)

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Operating System - The low-level software which handles the interface to peripheral hardware, schedules tasks, allocates storage, and presents a default interface to the user when no application program is running.

What is a Distributed System?

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- FOLDOC Definition:

  A collection of (probably heterogeneous) automata whose distribution is transparent to the user so that the system appears as one local machine. This is in contrast to a network, where the user is aware that there are several machines, and their location, storage replication, load balancing and functionality is not transparent. Distributed systems usually use some kind of client-server organization.

What is a Distributed System?

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Textbook Definitions:

A distributed system is a collection of independent computers that appear to the users of the system as a single computer.

[Andrew Tanenbaum]

A distributed system is several computers doing something together. Thus, a distributed system has three primary characteristics: multiple computers, interconnections, and shared state.

[Michael Schroeder]

Unsatisfactory

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Why are these definitions short?

Why do these definitions look inadequate to us?

Because we are interested in the insides of a distributed system

- Design and implementation
- Maintenance
- Algorithmics ("protocols" or "distributed algorithms")

The internet – Quick Refresher

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- Underlies many distributed systems.
- A vast interconnected collection of computer networks of many types.
- Intranets – subnetworks operated by companies and organizations.
- Intranets contain LANs (local area networks).
- WAN – wide area networks, consists of subnets (intranets, LANs, etc.)
- ISPs – Internet Service Providers. Companies that provide modem links and other types of connections to users.
- Intranets (actually the ISPs' core routers) are linked by backbones – network links of large bandwidth, such as satellite connections, fiber optic cables, and other high-bandwidth circuits.
- Google Fiber? (MAN = Metropolitan Area Networks)

An Intranet & a distributed system

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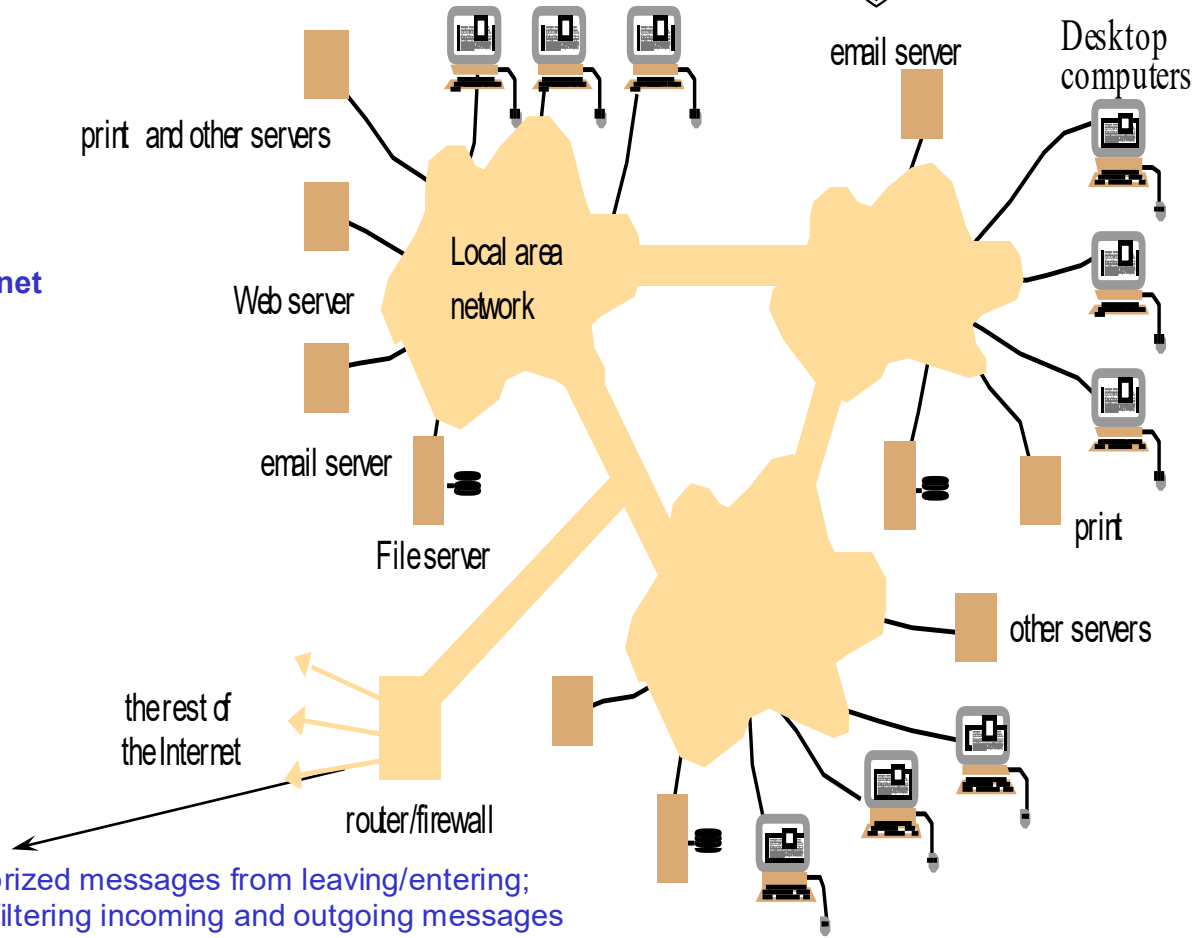
Running over this Intranet is a distributed file system



prevents unauthorized messages from leaving/entering; implemented by filtering incoming and outgoing messages via firewall “rules” (configurable)